HW03

ENAE 641: Linear System Dynamics

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Problem 1

The file `F16.mat` on ELMS contains a standard state-space perturbation model for the longitudinal dynamics of an F16 aircraft in level flight at 10000 ft moving with speed 500 ft s^{-1}. The four states of this model are: speed, angle of attack, pitch angle, and pitch rate, in that order.

a) Determine the real block modal form for the $\boldsymbol{A}$ matrix, and show that the system is stable.

b) Specify a physical (i.e. real-valued) initial condition which will uniquely excite the phugoid (slowly oscillating) mode. Repeat for the short period (faster oscillating) mode. Interpret each in terms of the physical variables.

c) The two inputs to this model are throttle and elevator deviations from trim, respectively. Which mode is most strongly excited by the throttle? Which mode is most strongly excited by the elevator?

HINT: Look at the corresponding modal $\boldsymbol{B}$ matrix for the transform in Part A

a) The eigenvalues of $\boldsymbol{A}$ form two complex-conjugate pairs:

$$\lambda_{1,2} = \sigma_1 \pm j\omega_1 = -0.01425 \pm 0.04434j, \quad \lambda_{3,4} = \sigma_3 \pm j\omega_3 = -1.695 \pm 2.058j$$

The slow pair is the phugoid ($\omega_n = 0.046\,58\,\text{rad}\,\text{s}^{-1}$, $\zeta = 0.306$, period $2\pi/\omega_1 \approx 141.7\,\text{s}$). The fast pair is the short period ($\omega_n = 2.666\,\text{rad}\,\text{s}^{-1}$, $\zeta = 0.636$, period $2\pi/\omega_3 \approx 3.05\,\text{s}$).

Let $\mathbf{v}_1$ and $\mathbf{v}_3$ be the eigenvectors for $\lambda_1 = \sigma_1 + j\omega_1$ and $\lambda_3 = \sigma_3 + j\omega_3$. Taking real and imaginary parts of $\boldsymbol{A}\mathbf{v}_1 = \lambda_1\mathbf{v}_1$ gives

$$\boldsymbol{A}[\text{Re}\{\mathbf{v}_1\}\ \ \text{Im}\{\mathbf{v}_1\}] = [\text{Re}\{\mathbf{v}_1\}\ \ \text{Im}\{\mathbf{v}_1\}]\begin{bmatrix}\sigma_1 & \omega_1\\ -\omega_1 & \sigma_1\end{bmatrix}$$

and likewise for $\mathbf{v}_3$. So the real transformation $\mathbf{x} = \boldsymbol{P}\mathbf{z}$ with

$$\boldsymbol{P} = [\text{Re}\{\mathbf{v}_1\}\ \ \text{Im}\{\mathbf{v}_1\}\ \ \text{Re}\{\mathbf{v}_3\}\ \ \text{Im}\{\mathbf{v}_3\}]$$

puts $\boldsymbol{A}_p = \boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P}$ in block modal form, with the phugoid in (z_1, z_2) and the short period in (z_3, z_4). Each eigenvector is scaled so that its largest entry is 1: $\mathbf{v}_1$ by its speed entry, and $\mathbf{v}_3$ by its pitch-rate entry. `MATLAB` then gives

$$\boldsymbol{P} = \begin{bmatrix} 1 & 0 & 0.01390 & -0.01653\\ -0.008132 & -0.0001973 & -0.06345 & -0.4306\\ -0.2367 & -0.6968 & -0.1014 & -0.1231\\ 0.08060 & -0.001325 & 1 & 0\end{bmatrix}$$

$$\boldsymbol{A}_p = \boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \begin{bmatrix} -0.01425 & 0.04434 & 0 & 0\\ -0.04434 & -0.01425 & 0 & 0\\ 0 & 0 & -1.695 & 2.058\\ 0 & 0 & -2.058 & -1.695\end{bmatrix} \quad \blacksquare$$

Stability: $\boldsymbol{A}_p$ is block diagonal, so with the rotation matrix $\boldsymbol{\Theta}(\theta) = \begin{bmatrix}\cos(\theta) & \sin(\theta)\\ -\sin(\theta) & \cos(\theta)\end{bmatrix}$,

$$e^{\boldsymbol{A}_p t} = \begin{bmatrix} e^{\sigma_1 t}\boldsymbol{\Theta}(\omega_1 t) & \mathbf{0}\\ \mathbf{0} & e^{\sigma_3 t}\boldsymbol{\Theta}(\omega_3 t)\end{bmatrix}, \quad \mathbf{x}(t) = \boldsymbol{P}e^{\boldsymbol{A}_p t}\boldsymbol{P}^{-1}\mathbf{x}_0$$

Each block is a rotation (norm 1) scaled by $e^{\sigma_k t}$, and $\sigma_3 < \sigma_1 < 0$, so $\left\|e^{\boldsymbol{A}_p t}\right\|_{i,2} = e^{\sigma_1 t}$ and

$$\|\mathbf{x}(t)\|_2 \le \|\boldsymbol{P}\|_{i,2}\left\|\boldsymbol{P}^{-1}\right\|_{i,2} e^{\sigma_1 t}\|\mathbf{x}_0\|_2 \to 0$$

for every $\mathbf{x}_0$. Both modes decay exponentially, i.e. all poles lie in the open left half-plane:

$$\max_k \mathrm{Re}(\lambda_k) = \sigma_1 = -0.01425 < 0 \quad \Rightarrow \quad \text{asymptotically (exponentially) stable} \quad \blacksquare$$

The `MATLAB` calculation:

```
eigenvalues of A:
       -1.6948 +     2.0576i
       -1.6948 -     2.0576i
      -0.014253 +   0.044344i
      -0.014253 -   0.044344i

modes (lambda = sigma +/- j omega):
          mode           sigma       omega      omega_n      zeta      period_s
    _____________    _________    ________    ________    _______    ________
    "phugoid"         -0.014253    0.044344    0.046579     0.306     141.69
    "short period"      -1.6948      2.0576      2.6657    0.63578     3.0537

max Re(lambda) = -0.0142529 < 0  ->  asymptotically stable

Pm = [Re(v_ph), Im(v_ph), Re(v_sp), Im(v_sp)]:
             1            0     0.013902    -0.016531
    -0.0081317  -0.00019725    -0.063454     -0.43063
      -0.23666     -0.69679     -0.10141     -0.12312
        0.0806   -0.0013248            1            0

A_p = Pm^-1 A Pm:
     -0.014253     0.044344            0            0
     -0.044344    -0.014253            0            0
             0            0      -1.6948       2.0576
             0            0      -2.0576      -1.6948
```

b) An initial state in a mode's plane excites only that mode. For $\mathbf{x}_0 = a\,\mathrm{Re}\{\mathbf{v}_1\} + b\,\mathrm{Im}\{\mathbf{v}_1\}$ with real a and b, $\mathbf{z}_0 = \boldsymbol{P}^{-1}\mathbf{x}_0 = [a, b, 0, 0]^\mathsf{T}$. Since $e^{\boldsymbol{A}_p t}$ is block diagonal, $z_3(t) = z_4(t) = 0$ for all t, and

$$\mathbf{x}(t) = e^{\sigma_1 t}[a\cos(\omega_1 t) + b\sin(\omega_1 t)]\,\mathrm{Re}\{\mathbf{v}_1\} + e^{\sigma_1 t}[b\cos(\omega_1 t) - a\sin(\omega_1 t)]\,\mathrm{Im}\{\mathbf{v}_1\}$$

contains only the phugoid and stays in the plane $\mathrm{span}\{\mathrm{Re}\{\mathbf{v}_1\}, \mathrm{Im}\{\mathbf{v}_1\}\}$. Any real a and b that are not both zero work. Since the model is linear, the initial state can be scaled down to stay a small perturbation from trim. Taking $a = 1$ and $b = 0$:

$$\mathbf{x}_0^{\mathrm{ph}} = \mathrm{Re}\{\mathbf{v}_1\} = \begin{bmatrix} 1 \\ -0.008132 \\ -0.2367 \\ 0.08060 \end{bmatrix} \quad \blacksquare$$

Likewise, $\mathbf{x}_0 = \mathrm{Re}\{\mathbf{v}_3\}$ gives $\mathbf{z}_0 = [0, 0, 1, 0]^\mathsf{T}$ and excites only the short period:

$$\mathbf{x}_0^{\text{sp}} = \text{Re}\{\mathbf{v}_3\} = \begin{bmatrix} 0.01390 \\ -0.06345 \\ -0.1014 \\ 1 \end{bmatrix} \quad \blacksquare$$

Interpretation: Write each eigenvector in polar form, $v_i = r_i e^{j\varphi_i}$. While a mode is excited, state x_i oscillates with relative amplitude r_i and relative phase φ_i:

state	phugoid r_i	phugoid φ_i	short period r_i	short period φ_i
speed	1	0°	0.0216	−49.9°
angle of attack	0.00813	−178.6°	0.435	−98.4°
pitch angle	0.736	−108.8°	0.160	−129.5°
pitch rate	0.0806	−0.9°	1	0°

The output matrix $\boldsymbol{C} = \left[\begin{smallmatrix} 500 & 0 & 0 & 0 \\ 0 & -500 & 500 & 0 \end{smallmatrix}\right]$ suggests that the speed state is normalized by $V_0 = 500\,\text{ft}\,\text{s}^{-1}$. Then $\mathbf{y}$ holds the speed change and the climb rate $V_0(\theta - \alpha)$, both in $\text{ft}\,\text{s}^{-1}$, and r_i compares a fractional speed change with α and θ in radians. The pitch-rate state has its own scaling: $\dot{\theta} = 0.4252\,q$ from the third row of $\boldsymbol{A}$. So its r_i is in those units, and measured as $\dot{\theta}$ the short period's pitch-rate amplitude is about 0.43, comparable to that of α.

- *Phugoid:* speed and pitch angle carry the motion. Pitch angle lags speed by about 109°, while the angle of attack barely moves ($r = 0.008$). With α nearly constant, the flight path angle $\theta - \alpha \approx \theta$, so the aircraft climbs as it slows down and dives as it speeds up. It trades kinetic for potential energy over a slow (about 142 s) oscillation at nearly constant angle of attack.
- *Short period:* pitch rate and angle of attack carry the motion. Angle of attack lags pitch rate by about 98°, and speed barely changes ($r = 0.02$). This is a fast (about 3 s), well-damped ($\zeta = 0.64$) pitching of the airframe about its center of mass at nearly constant speed.

The two initial states read the same way. $\mathbf{x}_0^{\text{ph}}$ starts the aircraft above trim speed with the nose down ($\theta = -0.237$ per unit of fractional speed), almost no change in α, and a small pitch rate. $\mathbf{x}_0^{\text{sp}}$ is essentially a pitch-rate kick, with small negative offsets in α and θ and almost no change in speed.

Simulating both initial states confirms that the other mode's modal coordinates stay at zero (to about 10^{-16}):

```
phugoid eigenvector v_ph (polar form, relative to speed):
            state              r        phi_deg
    ___________________    _________    ________
    "speed"                        1          0
    "angle of attack"    0.0081341     -178.61
    "pitch angle"          0.73588     -108.76
    "pitch rate"          0.080611    -0.94167

short-period eigenvector v_sp (polar form, relative to pitch rate):
            state              r       phi_deg
    ___________________    _______    _______
    "speed"                 0.0216    -49.936
    "angle of attack"      0.43528    -98.382
    "pitch angle"          0.15951    -129.48
    "pitch rate"                 1          0
```

```
16
17  phugoid x0 = Re(v_ph):
18              1   -0.0081317     -0.23666       0.0806
19  z0 = Pm^-1 x0:
20       1     0     0     0
21  max |z3|, |z4| over the phugoid response:  1.42e-16
22
23  short-period x0 = Re(v_sp):
24       0.013902    -0.063454     -0.10141            1
25  z0 = Pm^-1 x0:
26       0     0     1     0
27  max |z1|, |z2| over the short-period response:  3.72e-16
```

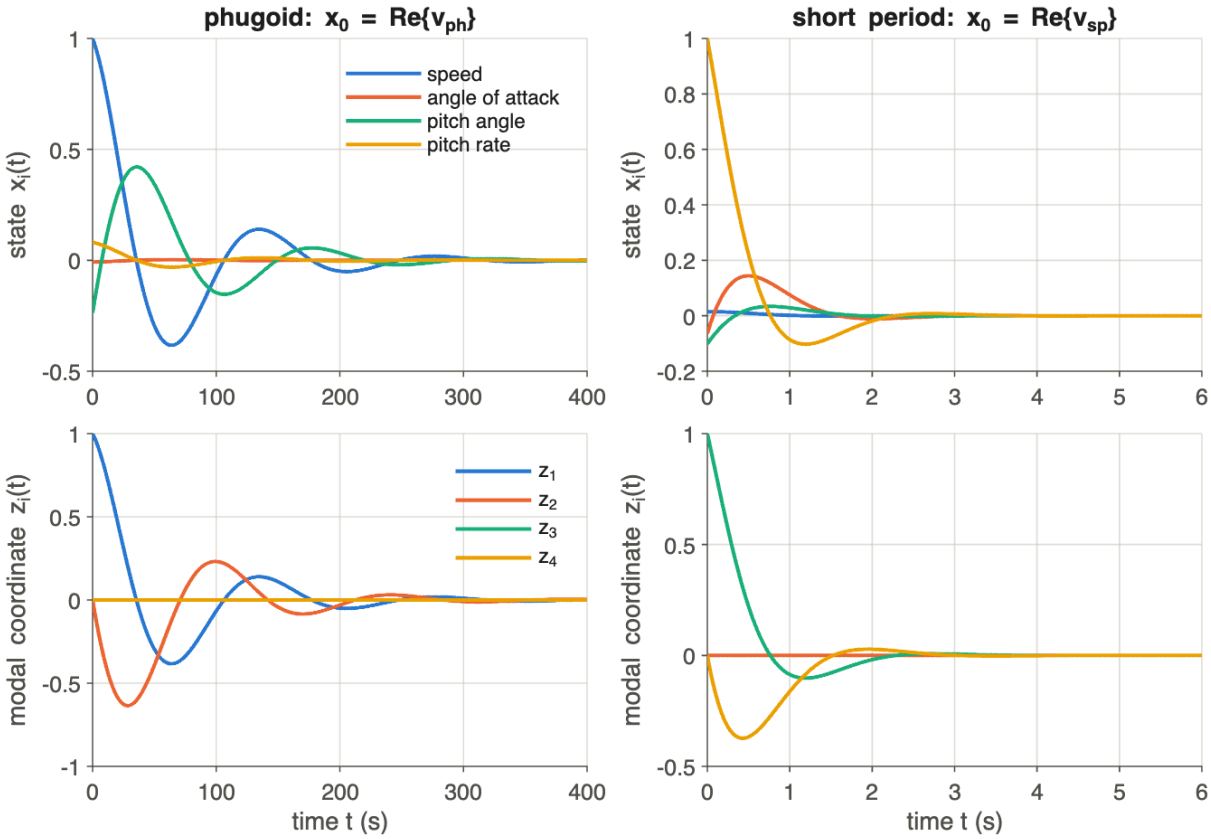


Figure 1: Homogeneous response (top) and modal coordinates $\mathbf{z} = \boldsymbol{P}^{-1}\mathbf{x}$ (bottom) for the phugoid initial state $\mathrm{Re}\{\mathbf{v}_1\}$ (left) and the short-period initial state $\mathrm{Re}\{\mathbf{v}_3\}$ (right).

c) In modal coordinates $\dot{\mathbf{z}} = \boldsymbol{A}_p\mathbf{z} + \boldsymbol{B}_p\mathbf{u}$ with $\boldsymbol{B}_p = \boldsymbol{P}^{-1}\boldsymbol{B}$. So entry (k, j) of $\boldsymbol{B}_p$ is how strongly input u_j drives z_k. Rows 1–2 drive the phugoid and rows 3–4 the short period; column 1 is throttle and column 2 is elevator:

$$\boldsymbol{B}_p = \boldsymbol{P}^{-1}\boldsymbol{B} = \begin{bmatrix} 0.05535 & 0.006327 \\ -0.01890 & 0.03786 \\ -0.004487 & -0.3446 \\ 0.004247 & 0.05742 \end{bmatrix} \quad \blacksquare$$

Comparing the size of each mode's pair of rows, for each input:

$$\text{throttle:}\quad \|[0.05535, -0.01890]\| = 0.0585 \quad \text{vs} \quad \|[-0.004487, 0.004247]\| = 0.00618 \quad (\approx 9.5\times)$$
$$\text{elevator:}\quad \|[0.006327, 0.03786]\| = 0.0384 \quad \text{vs} \quad \|[-0.3446, 0.05742]\| = 0.349 \quad (\approx 9.1\times)$$

with the phugoid block listed first in each line. The size of $\boldsymbol{B}_p$ depends on how the eigenvectors are scaled: rescaling $\mathbf{v}_k$ rescales mode k's rows inversely. A comparison that does not depend on that scaling is the part of each input direction $\mathbf{b}_j$ lying in mode k's plane, $\boldsymbol{P}_k\boldsymbol{B}_{p,k}\mathbf{e}_j$, where $\boldsymbol{P}_k$ holds mode k's two columns of $\boldsymbol{P}$ and $\boldsymbol{B}_{p,k}$ its two rows of $\boldsymbol{B}_p$. Its norm is 0.0555 vs 0.00475 for the throttle ($\approx$ 11.7 times) and 0.0286 vs 0.346 for the elevator ($\approx$ 12.1 times). Both measures agree:

$$\text{throttle} \Rightarrow \text{phugoid}, \quad \text{elevator} \Rightarrow \text{short period} \quad \blacksquare$$

This matches the physics, and the columns of $\boldsymbol{B}$ show it directly. The throttle changes thrust, which acts mainly on speed ($\boldsymbol{B}_{:,1} \approx 0.055\mathbf{e}_1$), and speed is the dominant component of the phugoid. The elevator produces a pitching moment, which acts mainly on pitch rate ($\boldsymbol{B}_{:,2} \approx -0.344\mathbf{e}_4$), and pitch rate and angle of attack are what move in the short period.

```
B_p = Pm^-1 B (columns: throttle, elevator):
      0.055354    0.0063272
     -0.018898     0.037862
    -0.0044866     -0.34461
     0.0042472     0.057417

modal-block norms ||B_p(block, j)||:
                     throttle    elevator
                     ________    ________
    phugoid          0.058491    0.038387
    short period     0.006178     0.34936
ratios (larger / smaller): throttle 9.47, elevator 9.1

projection of each input direction onto each mode's plane, ||Pm_k B_p,k(:, j)||:
                     throttle     elevator
                     _________    ________
    phugoid           0.055537    0.028592
    short period     0.0047472      0.3458
ratios (larger / smaller): throttle 11.7, elevator 12.1
```

Problem 2

Consider a time-periodic system from HW02:

$$\boldsymbol{A}(t) = \begin{bmatrix} -1 + 1.5\cos^2(t) & 1 - 1.5\sin(t)\cos(t) \\ -1 - 1.5\sin(t)\cos(t) & -1 + 1.5\sin^2(t) \end{bmatrix}$$

This system is time-periodic, hence it admits a Floquet decomposition. Determine the components $(\boldsymbol{P}(t), \boldsymbol{Q}, \boldsymbol{R})$ of this decomposition.

Every entry of $\boldsymbol{A}(t)$ is built from $\cos^2(t)$, $\sin(t)\cos(t)$, and $\sin^2(t)$, all of which have period π. So $\boldsymbol{A}(t+\pi) = \boldsymbol{A}(t)$, and the system is LTP with $T = \pi$. Take $t_0 = 0$.

State transition matrix: Define the rotation matrix $\boldsymbol{\Theta}(t)$ and the constant matrices $\boldsymbol{\Lambda}$ and $\boldsymbol{S}$:

$$\boldsymbol{\Theta}(t) = \begin{bmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{bmatrix}, \quad \boldsymbol{\Lambda} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -1 \end{bmatrix}, \quad \boldsymbol{S} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

These satisfy $\dot{\boldsymbol{\Theta}} = \boldsymbol{S}\boldsymbol{\Theta} = \boldsymbol{\Theta}\boldsymbol{S}$ and $\boldsymbol{\Theta}(t+\pi) = -\boldsymbol{\Theta}(t)$. Multiplying out (with $s = \sin(t)$, $c = \cos(t)$, and $c^2 + s^2 = 1$) shows that $\boldsymbol{A}(t)$ is a rotated copy of $\boldsymbol{\Lambda}$, plus $\boldsymbol{S}$:

$$\boldsymbol{\Theta}\boldsymbol{\Lambda}\boldsymbol{\Theta}^{\mathsf{T}} + \boldsymbol{S} = \begin{bmatrix} \frac{1}{2}c^2 - s^2 & -\frac{3}{2}sc \\ -\frac{3}{2}sc & \frac{1}{2}s^2 - c^2 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -1 + 1.5c^2 & 1 - 1.5sc \\ -1 - 1.5sc & -1 + 1.5s^2 \end{bmatrix} = \boldsymbol{A}(t)$$

Substituting $\mathbf{x} = \boldsymbol{\Theta}(t)\mathbf{w}$ into $\dot{\mathbf{x}} = \boldsymbol{A}(t)\mathbf{x}$ then gives a time-invariant system:

$$\dot{\mathbf{w}} = \boldsymbol{\Theta}^{\mathsf{T}}\left[\boldsymbol{A}\boldsymbol{\Theta} - \dot{\boldsymbol{\Theta}}\right]\mathbf{w} = \boldsymbol{\Theta}^{\mathsf{T}}[\boldsymbol{\Theta}\boldsymbol{\Lambda} + \boldsymbol{S}\boldsymbol{\Theta} - \boldsymbol{S}\boldsymbol{\Theta}]\mathbf{w} = \boldsymbol{\Lambda}\mathbf{w}$$

so the STM is

$$\boldsymbol{\Phi}(t, \tau) = \boldsymbol{\Theta}(t)e^{\boldsymbol{\Lambda}(t-\tau)}\boldsymbol{\Theta}^{\mathsf{T}}(\tau)$$

which agrees with the STM found in HW02.

Monodromy matrix: Since $\boldsymbol{\Theta}(\pi) = -\boldsymbol{I}$ and $\boldsymbol{\Theta}(0) = \boldsymbol{I}$,

$$\boldsymbol{Q} = \boldsymbol{\Phi}(t_0 + T, t_0) = \boldsymbol{\Phi}(\pi, 0) = -e^{\boldsymbol{\Lambda}\pi} = \begin{bmatrix} -e^{\pi/2} & 0 \\ 0 & -e^{-\pi} \end{bmatrix} \approx \begin{bmatrix} -4.810 & 0 \\ 0 & -0.0432 \end{bmatrix}$$

Constant matrix $\boldsymbol{R}$: $\boldsymbol{R}$ is defined by $e^{\boldsymbol{R}T} = \boldsymbol{Q}$, so $\boldsymbol{R} = \frac{1}{T}\log\boldsymbol{Q}$. No real $\boldsymbol{R}$ works here. The eigenvalues of $\boldsymbol{Q}$ are $e^{\pi r_k}$, where r_k are the eigenvalues of $\boldsymbol{R}$. Making $e^{\pi r_k}$ negative requires $\mathrm{Im}(r_k)$ to be an odd integer, so r_k is not real. A real $\boldsymbol{R}$ would then have the eigenvalue pair r_k, $\bar{r}_k$, whose exponentials are equal. That would make the eigenvalues of $\boldsymbol{Q}$ equal, but $-e^{\pi/2} \neq -e^{-\pi}$. The principal matrix logarithm (`MATLAB`'s `logm()`), with $\log(-a) = \ln(a) + j\pi$, gives

$$\boldsymbol{R} = \frac{1}{\pi}\log\boldsymbol{Q} = \frac{1}{\pi}\begin{bmatrix} \pi/2 + j\pi & 0 \\ 0 & -\pi + j\pi \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + j & 0 \\ 0 & -1 + j \end{bmatrix} = \boldsymbol{\Lambda} + j\boldsymbol{I}$$

Periodic part $\boldsymbol{P}(t)$: Since $\boldsymbol{\Lambda}$ and $j\boldsymbol{I}$ commute, $e^{-\boldsymbol{R}t} = e^{-\boldsymbol{\Lambda}t}e^{-jt}$, and

$$\boldsymbol{P}(t) = \boldsymbol{\Phi}(t, t_0)e^{-\boldsymbol{R}t} = \boldsymbol{\Theta}(t)e^{\boldsymbol{\Lambda}t}e^{-\boldsymbol{\Lambda}t}e^{-jt} = e^{-jt}\boldsymbol{\Theta}(t), \quad \boldsymbol{P}^{-1}(t) = e^{jt}\boldsymbol{\Theta}^{\mathsf{T}}(t)$$

This $\boldsymbol{P}(t)$ has the properties the decomposition requires:

- It is periodic: $\boldsymbol{P}(t+\pi) = e^{-jt}e^{-j\pi}\boldsymbol{\Theta}(t+\pi) = e^{-jt}(-1)(-\boldsymbol{\Theta}(t)) = \boldsymbol{P}(t)$.

- It is a Lyapunov transformation: $|\det \boldsymbol{P}(t)| = \left|e^{-2jt}\right| \det \boldsymbol{\Theta}(t) = 1$ and $\left|\det \boldsymbol{P}^{-1}(t)\right| = 1$ for all t, so both are bounded away from zero.
- It reproduces the STM: $\boldsymbol{P}(t)e^{\boldsymbol{R}(t-\tau)}\boldsymbol{P}^{-1}(\tau) = e^{-jt}\boldsymbol{\Theta}(t)e^{\boldsymbol{\Lambda}(t-\tau)}e^{j(t-\tau)}e^{j\tau}\boldsymbol{\Theta}^{\mathsf{T}}(\tau) = \boldsymbol{\Phi}(t,\tau)$.

So the Floquet decomposition $\boldsymbol{\Phi}(t,\tau) = \boldsymbol{P}(t)e^{\boldsymbol{R}(t-\tau)}\boldsymbol{P}^{-1}(\tau)$, with period $T = \pi$, has

$$\boldsymbol{P}(t) = e^{-jt}\begin{bmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{bmatrix}, \quad \boldsymbol{Q} = \begin{bmatrix} -e^{\pi/2} & 0 \\ 0 & -e^{-\pi} \end{bmatrix}, \quad \boldsymbol{R} = \begin{bmatrix} \frac{1}{2} + j & 0 \\ 0 & -1 + j \end{bmatrix} \quad \blacksquare$$

A real-valued alternative: $\boldsymbol{A}(t)$ is also 2π-periodic. Over $T = 2\pi$ the monodromy matrix has positive eigenvalues, so it has a real logarithm and every component is real:

$$\boldsymbol{Q} = \boldsymbol{\Phi}(2\pi, 0) = e^{2\pi\boldsymbol{\Lambda}} = \begin{bmatrix} e^{\pi} & 0 \\ 0 & e^{-2\pi} \end{bmatrix}, \quad \boldsymbol{R} = \frac{1}{2\pi}\log \boldsymbol{Q} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -1 \end{bmatrix}, \quad \boldsymbol{P}(t) = \boldsymbol{\Theta}(t) = \begin{bmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{bmatrix}$$

In either case $\boldsymbol{R}$ is not unique: other branches of the logarithm give $\boldsymbol{R} = \operatorname{diag}\left(\frac{1}{2} + (2k+1)j, -1 + (2m+1)j\right)$ for $T = \pi$ and $\boldsymbol{R} = \operatorname{diag}\left(\frac{1}{2} + kj, -1 + mj\right)$ for $T = 2\pi$ (k, m integers), each with its own $\boldsymbol{P}(t)$. For a given T, the Floquet multipliers ($\operatorname{eig}(\boldsymbol{Q})$) are unique. For every choice, the real parts of the Floquet exponents (the eigenvalues of $\boldsymbol{R}$) are $\frac{1}{2}$ and -1.

Stability: In the coordinates $\mathbf{z} = \boldsymbol{P}^{-1}(t)\mathbf{x}$ the dynamics are $\dot{\mathbf{z}} = \boldsymbol{R}\mathbf{z}$. A Lyapunov transformation preserves convergence and divergence. $\boldsymbol{R}$ has an eigenvalue with $\operatorname{Re}(r_1) = \frac{1}{2} > 0$, equivalently a Floquet multiplier with $\left|-e^{\pi/2}\right| \approx 4.81 > 1$, so the system is unstable. This is consistent with HW02, where almost every solution grows like $e^{t/2}$.

Problem 3

For the linear time-varying system

$$\dot{\mathbf{x}}(t) = \boldsymbol{A}(t)\mathbf{x}(t) \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

if

$$\boldsymbol{A}(t) = \begin{bmatrix} 0 & t\sin(t)\cos\left(\sqrt{t}\right) \\ -t\sin(t)\cos\left(\sqrt{t}\right) & 0 \end{bmatrix}$$

and $\mathbf{x}_0^\mathsf{T} = [-1, 2]$, find $\|\mathbf{x}(t)\|_2$ for all $t \geq t_0$. What does this tell you about the stability properties of this system (unstable, asymptotically stable, or stable)?

The system matrix is a scalar function times a fixed skew-symmetric matrix:

$$\boldsymbol{A}(t) = a(t)\boldsymbol{S}, \quad a(t) = t\sin(t)\cos\left(\sqrt{t}\right), \quad \boldsymbol{S} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \boldsymbol{S}^\mathsf{T} = -\boldsymbol{S}$$

Here $a(t)$ is real for every t: for $t < 0$, $\cos\left(\sqrt{t}\right) = \cosh\left(\sqrt{|t|}\right)$. So the argument below holds for any t_0. Take $V(\mathbf{x}) = \|\mathbf{x}\|_2^2$, i.e. $\boldsymbol{P} = \boldsymbol{I}$, so that $\boldsymbol{Q}(t) = -[\boldsymbol{A}^\mathsf{T}(t) + \boldsymbol{A}(t)]$. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\mathbf{x}(t)\|_2^2 = \mathbf{x}^\mathsf{T}(t)[\boldsymbol{A}^\mathsf{T}(t) + \boldsymbol{A}(t)]\mathbf{x}(t) = a(t)\mathbf{x}^\mathsf{T}(t)[\boldsymbol{S}^\mathsf{T} + \boldsymbol{S}]\mathbf{x}(t) = 0$$

so the norm never changes from its initial value:

$$\|\mathbf{x}(t)\|_2 = \|\mathbf{x}_0\|_2 = \sqrt{(-1)^2 + 2^2} = \sqrt{5} \approx 2.236 \quad \forall t \geq t_0 \quad \blacksquare$$

The solution itself is a rotation. Since $\boldsymbol{A}(t)\boldsymbol{A}(\tau) = a(t)a(\tau)\boldsymbol{S}^2 = \boldsymbol{A}(\tau)\boldsymbol{A}(t)$, the STM is the exponential of the integrated system matrix (the commuting case from HW02). With $\boldsymbol{S}^2 = -\boldsymbol{I}$,

$$\boldsymbol{\Phi}(t, t_0) = e^{\theta \boldsymbol{S}} = \cos(\theta)\boldsymbol{I} + \sin(\theta)\boldsymbol{S} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}, \quad \theta(t, t_0) = \int_{t_0}^{t} \sigma\sin(\sigma)\cos\left(\sqrt{\sigma}\right)\mathrm{d}\sigma$$

So $\mathbf{x}(t) = \boldsymbol{\Phi}(t, t_0)\mathbf{x}_0$ is $\mathbf{x}_0$ rotated by the angle $\theta(t, t_0)$. This angle has no elementary closed form, but it is not needed for the norm.

Stability: The argument above holds for every $\mathbf{x}_0$ and every t_0, so $\|\mathbf{x}(t)\|_2 = \|\mathbf{x}_0\|_2$, i.e. $\|\boldsymbol{\Phi}(t, t_0)\|_{i,2} = 1$:

- *Stable isL:* $\|\mathbf{x}(t)\|_2 \leq \gamma\|\mathbf{x}_0\|_2$ with $\gamma = 1$, independent of t_0, so the stability is uniform. This matches the lecture test: $\boldsymbol{P} = \boldsymbol{I}$ gives $\boldsymbol{Q} = \mathbf{0}$, which is positive semi-definite.
- *Not asymptotically stable:* $\|\mathbf{x}(t)\|_2 = \|\mathbf{x}_0\|_2 \not\to 0$ for any $\mathbf{x}_0 \neq \mathbf{0}$, so it is not exponentially stable either.
- *Not unstable:* every solution stays bounded.

The state keeps circling at radius $\sqrt{5}$ about the origin. It never escapes and never decays:

$$\|\mathbf{x}(t)\|_2 = \sqrt{5} \quad \Rightarrow \quad \text{stable isL (uniformly), but not asymptotically stable} \quad \blacksquare$$

Problem 4

a) For the F16 model in Problem 1, find a constant, positive-definite matrix $\boldsymbol{P}$ so that

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\mathbf{x}^{\mathsf{T}}(t)\boldsymbol{P}\mathbf{x}(t)\Big) = -\|\mathbf{x}(t)\|^2$$

(Use `MATLAB`'s `lyap()` function, but read its documentation carefully!) Verify this $\boldsymbol{P}$ is positive definite.

b) Use this $(\boldsymbol{P}, \boldsymbol{Q})$ pair to bound the exponential convergence of $\|\mathbf{x}(t)\|$ from an arbitrary $\mathbf{x}_0$.

a) For a constant $\boldsymbol{P}$ and $\dot{\mathbf{x}} = \boldsymbol{A}\mathbf{x}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{x}^{\mathsf{T}}\boldsymbol{P}\mathbf{x}) = \dot{\mathbf{x}}^{\mathsf{T}}\boldsymbol{P}\mathbf{x} + \mathbf{x}^{\mathsf{T}}\boldsymbol{P}\dot{\mathbf{x}} = \mathbf{x}^{\mathsf{T}}(\boldsymbol{A}^{\mathsf{T}}\boldsymbol{P} + \boldsymbol{P}\boldsymbol{A})\mathbf{x}$$

This equals $-\|\mathbf{x}\|_2^2 = -\mathbf{x}^{\mathsf{T}}\boldsymbol{I}\mathbf{x}$ for every $\mathbf{x}$ exactly when $\boldsymbol{P}$ solves the algebraic Lyapunov equation with $\boldsymbol{Q} = \boldsymbol{I}$:

$$\boldsymbol{A}^{\mathsf{T}}\boldsymbol{P} + \boldsymbol{P}\boldsymbol{A} = -\boldsymbol{I}$$

`MATLAB`'s `lyap(A, Q)` solves $\boldsymbol{A}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{A}^{\mathsf{T}} + \boldsymbol{Q} = \boldsymbol{0}$, which has the transpose on the other factor. Calling `lyap(A, eye(4))` would therefore solve the wrong equation. That mistake would not show up in a positive-definiteness check either. Its solution is also positive definite, since $\boldsymbol{A}^{\mathsf{T}}$ is Hurwitz too. Passing $\boldsymbol{A}^{\mathsf{T}}$ instead, `P = lyap(A', eye(4))` solves $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{P} + \boldsymbol{P}\boldsymbol{A} + \boldsymbol{I} = \boldsymbol{0}$ and gives

$$\boldsymbol{P} = \begin{bmatrix} 27.17 & -3.394 & 8.074 & 0.2346 \\ -3.394 & 5.573 & -16.59 & -1.434 \\ 8.074 & -16.59 & 57.44 & 4.671 \\ 0.2346 & -1.434 & 4.671 & 0.5930 \end{bmatrix} \quad \blacksquare$$

Positive definiteness: $\boldsymbol{P}$ is symmetric, and all four of its eigenvalues are positive (a Cholesky factorization also succeeds). By Rayleigh–Ritz, $\mathbf{x}^{\mathsf{T}}\boldsymbol{P}\mathbf{x} \geq \lambda_{\min}(\boldsymbol{P})\|\mathbf{x}\|_2^2 > 0$ for every $\mathbf{x} \neq \boldsymbol{0}$:

$$\lambda(\boldsymbol{P}) = \{0.1873, 0.6929, 25.20, 64.69\} \quad \Rightarrow \quad \lambda_{\min}(\boldsymbol{P}) = 0.1873 > 0 \quad \Rightarrow \quad \boldsymbol{P} \text{ is positive definite} \quad \blacksquare$$

```
P = lyap(A', I):
        27.165      -3.3937       8.0743      0.23456
       -3.3937       5.5732      -16.594      -1.4335
        8.0743      -16.594       57.437       4.6712
       0.23456      -1.4335       4.6712      0.59301

eigenvalues of P:
       0.18734      0.69286       25.196       64.692
min eigenvalue = 0.18734 > 0,  chol flag = 0  ->  P is positive definite
```

b) Use the pair $(\boldsymbol{P}, \boldsymbol{Q}) = (\boldsymbol{P}, \boldsymbol{I})$ from Part A, with

$$\rho = \lambda_{\max}(\boldsymbol{P}) = 64.69, \quad \eta = \lambda_{\min}(\boldsymbol{P}) = 0.1873, \quad \nu = \lambda_{\min}(\boldsymbol{Q}) = 1$$

With $V = \mathbf{x}^{\mathsf{T}}\boldsymbol{P}\mathbf{x}$, Rayleigh–Ritz gives $\eta\|\mathbf{x}\|_2^2 \leq V \leq \rho\|\mathbf{x}\|_2^2$. Since $\nu > 0$,

$$\dot{V} = -\mathbf{x}^{\mathsf{T}}\boldsymbol{Q}\mathbf{x} \leq -\nu\|\mathbf{x}\|_2^2 \leq -\frac{\nu}{\rho}V \quad \Rightarrow \quad V(t) \leq V(t_0)e^{-(\nu/\rho)(t-t_0)}$$

by the comparison lemma. Then $\eta\|\mathbf{x}(t)\|_2^2 \le V(t)$ and $V(t_0) \le \rho\|\mathbf{x}_0\|_2^2$ give the lecture bound

$$\|\mathbf{x}(t)\|_2 \le \sqrt{\frac{\rho}{\eta}} e^{-\lambda(t-t_0)} \|\mathbf{x}_0\|_2, \quad \gamma = \sqrt{\frac{\rho}{\eta}} = 18.58, \quad \lambda = \frac{\nu}{2\rho} = 0.007\,729 \text{ s}^{-1}$$

which holds from an arbitrary $\mathbf{x}_0$ and t_0:

$$\|\mathbf{x}(t)\|_2 \le 18.58 e^{-0.007729(t-t_0)} \|\mathbf{x}_0\|_2 \quad \blacksquare$$

This is a valid but conservative bound:

- The guaranteed rate $\lambda = 0.007\,73$ s^{-1} is about half the true slowest decay rate $|\sigma_1| = 0.014\,25$ s^{-1} from Problem 1a. Since $\lambda = \nu/(2\rho)$, this gap depends on ρ alone, not on η. ρ is set mainly by the phugoid. Restricted to the phugoid's plane, $\boldsymbol{P}$ already has largest eigenvalue ≈ 59.7, above the $1/(2|\sigma_1|) = 35.1$ that a mode with orthogonal, equal-length $\text{Re}\{\mathbf{v}_1\}$ and $\text{Im}\{\mathbf{v}_1\}$ would give. Here those vectors have lengths 1.03 and 0.70 and are 77° apart, so the phugoid is not "round" in these coordinates.
- The overshoot factor $\gamma = 18.58$ is far above the true worst-case gain $\max_t \left\|e^{\boldsymbol{A}t}\right\|_{i,2} = 1.127$, which is reached at $t \approx 17.5$ s. Even at the guaranteed rate, the smallest valid constant is $\max_t \left\|e^{\boldsymbol{A}t}\right\|_{i,2} e^{\lambda t} = 1.34$. $\gamma = \sqrt{\rho/\eta}$ is large because the directions of ρ and η come from different modes. ρ comes mostly from the slow phugoid, while the eigenvector for η lies in the fast short-period plane, and their decay rates differ by a factor of about $|\sigma_3|/|\sigma_1| \approx 119$.
- For comparison, let $\boldsymbol{P}_m$ be the modal transformation of Problem 1a (not the Lyapunov $\boldsymbol{P}$ above). It gives the much tighter $\|\mathbf{x}(t)\|_2 \le \|\boldsymbol{P}_m\|_{i,2} \left\|\boldsymbol{P}_m^{-1}\right\|_{i,2} e^{\sigma_1 t} \|\mathbf{x}_0\|_2 = 2.62 e^{-0.01425t} \|\mathbf{x}_0\|_2$. The Lyapunov bound is looser, but it needs no eigenvectors, and the same argument carries over to a time-varying $\boldsymbol{A}(t)$.

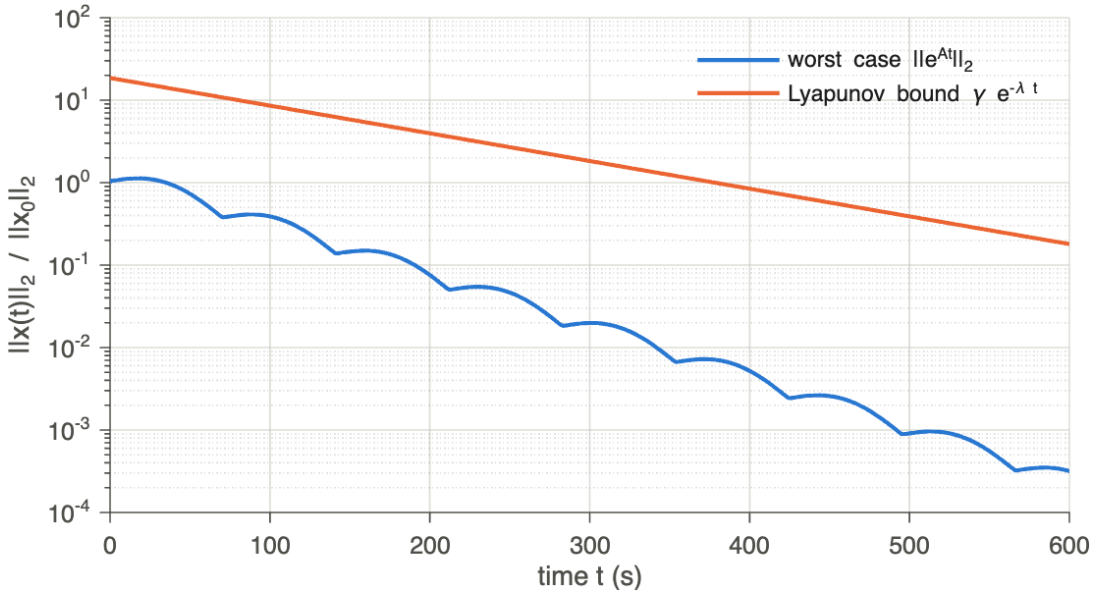


Figure 2: Worst-case normalized response $\left\|e^{\boldsymbol{A}t}\right\|_{i,2} = \max_{\mathbf{x}_0} \|\mathbf{x}(t)\|_2/\|\mathbf{x}_0\|_2$ against the Lyapunov bound $\gamma e^{-\lambda t}$.

```
rho = lambda_max(P) = 64.6918
eta = lambda_min(P) = 0.18734
nu  = lambda_min(Q) = 1
gamma  = sqrt(rho / eta) = 18.5827
```

```
lambda = nu / (2 rho)   = 0.00772896 1/s

slowest decay rate of A, min |Re(lambda)| = 0.0142529 1/s
max_t ||e^(At)||_2 = 1.12748 at t = 17.49 s
bound >= ||e^(At)||_2 on the whole grid: true

lambda_max of P restricted to the phugoid plane = 59.69  (1 / (2 |sigma_1|) = 35.08)
|Re(v_ph)| = 1.03, |Im(v_ph)| = 0.697, angle between them = 76.7 deg
smallest M with ||e^(At)||_2 <= M e^(-lambda t): 1.342
|sigma_3| / |sigma_1| = 118.9
modal bound: ||x(t)|| <= cond(Pm) e^(sigma_1 t) ||x0||, cond(Pm) = 2.621
```

Problem 5

a) For the F16 model in Problem 1, what are the eigenvalues of $\boldsymbol{A} + \boldsymbol{A}^\mathsf{T}$? Do these eigenvalues allow you to conclude the kind of exponential stability result that you obtained in Problem 1 and Problem 4?

b) Is there a contradiction between the results of Part A and that of Problem 1 and Problem 4? Explain carefully in light of Lyapunov theory.

HINT: What $\boldsymbol{P}$ are you implicitly using in the test of Part A? Should you expect an arbitrary choice such as this to satisfy Lyapunov's requirement?

a) `MATLAB` gives

$$\lambda(\boldsymbol{A} + \boldsymbol{A}^\mathsf{T}) = \{-7.327, -0.2504, 0.04021, 0.7006\} \quad \blacksquare$$

These are the eigenvalues in the test from lecture: along solutions,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\mathbf{x}\|_2^2 = \mathbf{x}^\mathsf{T}(\boldsymbol{A}^\mathsf{T} + \boldsymbol{A})\mathbf{x}$$

The theorem says the system is stable isL if $\lambda_{\max}(\boldsymbol{A}^\mathsf{T} + \boldsymbol{A}) \leq 0$. If instead $\boldsymbol{A}^\mathsf{T} + \boldsymbol{A} \leq -\varepsilon \boldsymbol{I}$, i.e. $\boldsymbol{Q} = -(\boldsymbol{A}^\mathsf{T} + \boldsymbol{A})$ is uniformly positive definite, it is exponentially stable. Here two eigenvalues are positive, so $\boldsymbol{Q}$ is indefinite. Neither condition holds, and the test cannot give exponential stability, or even stability isL. One entry already rules out the exponential test: $a_{33} = 0$, so $\mathbf{e}_3^\mathsf{T}(\boldsymbol{A} + \boldsymbol{A}^\mathsf{T})\mathbf{e}_3 = 2a_{33} = 0$, and $\boldsymbol{A} + \boldsymbol{A}^\mathsf{T}$ cannot be negative definite. Along the eigenvector $\mathbf{w}$ of the largest eigenvalue, $\|\mathbf{x}\|_2^2$ actually starts out increasing: $\frac{\mathrm{d}}{\mathrm{d}t}\|\mathbf{x}\|_2^2 = 0.7006\|\mathbf{x}_0\|_2^2 > 0$ at $\mathbf{x}_0 = \mathbf{w}$ (Part B).

$$\lambda_{\max}(\boldsymbol{A} + \boldsymbol{A}^\mathsf{T}) = 0.7006 > 0 \quad \Rightarrow \quad \text{no: the test is inconclusive} \quad \blacksquare$$

```
1 eigenvalues of A + A':
2       -7.3267     -0.25043      0.04021      0.70064
3 number of positive eigenvalues: 2
```

b) There is no contradiction. The test in Part A is Lyapunov's test with the particular choice $\boldsymbol{P} = \boldsymbol{I}$: $V = \mathbf{x}^\mathsf{T}\boldsymbol{I}\mathbf{x} = \|\mathbf{x}\|_2^2$, and $\boldsymbol{Q} = -(\boldsymbol{A}^\mathsf{T}\boldsymbol{I} + \boldsymbol{I}\boldsymbol{A}) = -(\boldsymbol{A} + \boldsymbol{A}^\mathsf{T})$. Lyapunov's theorem is only a sufficient condition. When a particular $(\boldsymbol{P}, \boldsymbol{Q})$ pair fails, the result is inconclusive, not a proof of instability, and a different $\boldsymbol{P}$ may succeed.

The converse theorem says what to expect. $\boldsymbol{A}$ is Hurwitz (Problem 1a), so for *any* positive definite $\boldsymbol{Q}$ there exists a positive definite $\boldsymbol{P}$ with $\boldsymbol{A}^\mathsf{T}\boldsymbol{P} + \boldsymbol{P}\boldsymbol{A} = -\boldsymbol{Q}$. Problem 4 found that $\boldsymbol{P}$ for $\boldsymbol{Q} = \boldsymbol{I}$. The theorem does not say the reverse, that an arbitrary positive definite $\boldsymbol{P}$ yields a positive definite $\boldsymbol{Q}$. Choosing $\boldsymbol{P} = \boldsymbol{I}$ succeeds only if $\boldsymbol{A} + \boldsymbol{A}^\mathsf{T}$ is negative definite, i.e. the Euclidean norm $\|\mathbf{x}\|_2$ shrinks along *every* solution at *every* instant. That is much stronger than all eigenvalues lying in the left half-plane, and this $\boldsymbol{A}$ does not satisfy it. $\boldsymbol{A}$ is far from normal: its couplings are lopsided (e.g. $a_{42} = -4.78$ but $a_{24} = 0.907$) and its states have very different scales. So the spherical level sets of $\|\mathbf{x}\|_2^2$ are not adapted to the dynamics. The ellipsoidal level sets of $\mathbf{x}^\mathsf{T}\boldsymbol{P}\mathbf{x}$ from Problem 4 are.

Simulation shows that $\|\mathbf{x}\|_2$ really does grow for a while from some initial states:

- From $\mathbf{x}_0 = \mathbf{w}$ (the top eigenvector of $\boldsymbol{A} + \boldsymbol{A}^\mathsf{T}$), $\|\mathbf{x}\|_2$ rises to $1.035\|\mathbf{x}_0\|_2$ at $t \approx 0.16\,\mathrm{s}$.
- From the worst-case unit initial state $\mathbf{x}_w$ for $t = 17.5\,\mathrm{s}$, it rises to $1.127\|\mathbf{x}_0\|_2$.

So $\|\mathbf{x}\|_2^2$ is not nonincreasing along all solutions, and no argument based on $\boldsymbol{P} = \boldsymbol{I}$ alone could certify the decay. Along the same solutions $V = \mathbf{x}^\mathsf{T}\boldsymbol{P}\mathbf{x}$ decreases monotonically ($\dot{V} = -\|\mathbf{x}\|_2^2 < 0$). Both solutions are still decaying at the end of the plot, and they converge to zero, as Problem 1a and

Problem 4b guarantee. An arbitrary $\boldsymbol{P}$ such as $\boldsymbol{I}$ should not be expected to satisfy Lyapunov's requirement. The part (a) test simply used the wrong $\boldsymbol{P}$ for this $\boldsymbol{A}$:

$$\boldsymbol{P} = \boldsymbol{I} \Rightarrow \boldsymbol{Q} \text{ indefinite (inconclusive)}, \quad \boldsymbol{P} = \text{lyap}(\boldsymbol{A}^{\mathsf{T}}, \boldsymbol{I}) \Rightarrow \boldsymbol{Q} = \boldsymbol{I} \succ 0 \quad \Rightarrow \quad \text{no contradiction} \quad \blacksquare$$

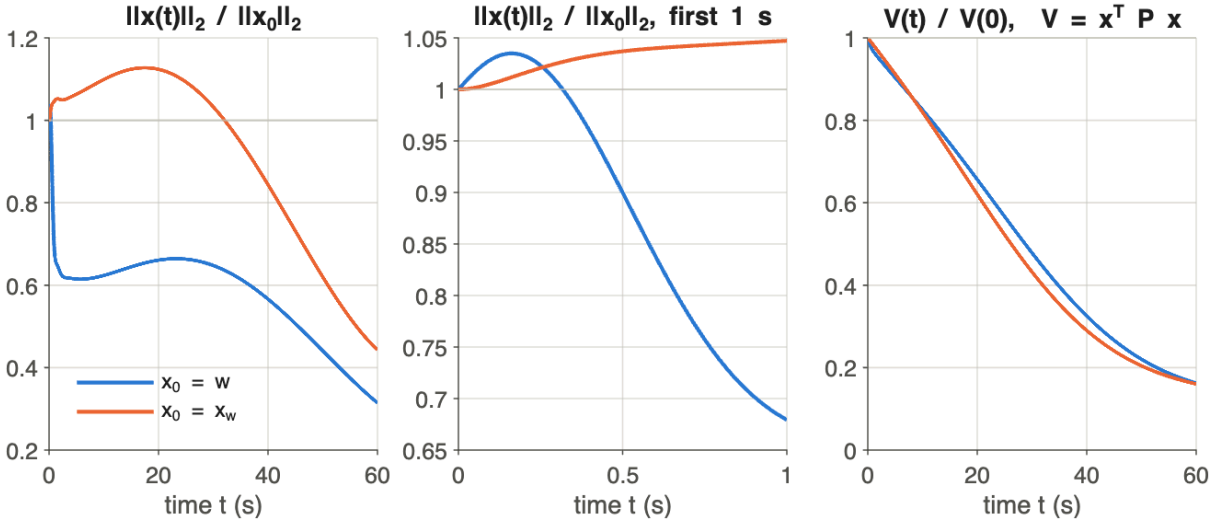


Figure 3: Euclidean norm over 60 s (left) and over the first second (middle), and $V = \mathbf{x}^{\mathsf{T}}\boldsymbol{P}\mathbf{x}$ (right), each normalized by its initial value, from $\mathbf{x}_0 = \mathbf{w}$ and from the worst-case initial state $\mathbf{x}_w$.

```
x0 = w, the unit eigenvector of A + A' for its largest eigenvalue:
         0.241     -0.68199      0.34272      0.59946
d/dt ||x||^2 at t = 0: w'(A + A')w = 0.700643 > 0
max ||x(t)|| / ||x0|| = 1.03493 at t = 0.16 s (back below 1 at t = 0.32 s)

x0 = x_w, the unit vector maximizing ||e^(At)x0|| at t = 17.49 s:
     -0.015632      0.27454     -0.95815     -0.07953
max ||x(t)|| / ||x0|| = 1.12748 at t = 17.49 s

V = x'Px nonincreasing along both solutions: true
```

Problem 6

For the homogenous dynamics of a linear system, suppose that the symmetric matrices $\boldsymbol{P}(t), \boldsymbol{Q}(t)$ satisfy the Lyapunov equation, with $\boldsymbol{P}(t)$ bounded and uniformly positive definite, and $\boldsymbol{Q}(t)$ bounded. Let $\nu(t)$ be the minimal eigenvalue of $\boldsymbol{Q}(t)$, i.e.

$$\nu(t) = \min_k \lambda_k(\boldsymbol{Q}(t))$$

Determine the "strongest" property (in order: inconclusive, stable, asymptotically stable, or exponentially stable) you can conclude for the following cases below. Show your reasoning carefully, and note that some or all of these cases might not fit neatly into the categories specifically considered in class.

a) $\nu(t) = \frac{1}{t}$

b) $\nu(t) = -ce^{-t}$ with $c > 0$

c) $\nu(t) = \frac{1}{t^2}$

d) $\nu(t) = 2 - 3\sin(t)$

a) The same bound is used in all four parts. Let $V(\mathbf{x}, t) = \mathbf{x}^\mathsf{T}\boldsymbol{P}(t)\mathbf{x}$, with $\rho = \max_t \lambda_{\max}(\boldsymbol{P}(t)) < \infty$ and $\eta = \min_t \lambda_{\min}(\boldsymbol{P}(t)) > 0$. By Rayleigh–Ritz,

$$\eta\|\mathbf{x}\|_2^2 \le V \le \rho\|\mathbf{x}\|_2^2$$

and along solutions, by the Lyapunov equation $-\boldsymbol{Q} = \dot{\boldsymbol{P}} + \boldsymbol{A}^\mathsf{T}\boldsymbol{P} + \boldsymbol{P}\boldsymbol{A}$,

$$\dot{V} = -\mathbf{x}^\mathsf{T}\boldsymbol{Q}(t)\mathbf{x} \le -\nu(t)\|\mathbf{x}\|_2^2$$

To compare $\dot{V}$ with V, the sign of $\nu(t)$ decides which side of the Rayleigh–Ritz bound applies:

- Where $\nu(t) \ge 0$, use $\|\mathbf{x}\|_2^2 \ge V/\rho$: $\quad -\nu\|\mathbf{x}\|_2^2 \le -(\nu/\rho)V$.
- Where $\nu(t) < 0$, use $\|\mathbf{x}\|_2^2 \le V/\eta$: $\quad -\nu\|\mathbf{x}\|_2^2 = |\nu|\|\mathbf{x}\|_2^2 \le (|\nu|/\eta)V$.

Both cases are covered by

$$\dot{V} \le -\beta(t)V, \quad \beta(t) = \min\left\{\frac{\nu(t)}{\rho}, \frac{\nu(t)}{\eta}\right\} = \frac{\nu_+(t)}{\rho} - \frac{\nu_-(t)}{\eta}, \quad \nu_\pm = \max\{\pm\nu, 0\}$$

Comparing with $\dot{w} = -\beta(t)w$, $w(t_0) = V(t_0)$ (comparison lemma) gives $V(t) \le V(t_0)e^{-\int_{t_0}^t \beta(\sigma)\,\mathrm{d}\sigma}$. Then

$$\|\mathbf{x}(t)\|_2 \le \sqrt{\frac{\rho}{\eta}}\exp\left(-\frac{1}{2}\int_{t_0}^t \beta(\sigma)\,\mathrm{d}\sigma\right)\|\mathbf{x}_0\|_2 \quad (*)$$

so each case comes down to the behavior of $\int_{t_0}^t \beta$.

To show a conclusion cannot be strengthened, I use the scalar test system $\dot{x} = -\frac{1}{2}\nu(t)x$ with $P = 1$ ($\rho = \eta = 1$). Its Lyapunov equation gives $Q = -2\left(-\frac{1}{2}\nu\right) = \nu(t)$ exactly. Its solution $x(t) = x_0\exp\left(-\frac{1}{2}\int_{t_0}^t \nu\right)$ makes $(*)$ an equality, and it satisfies every hypothesis whenever $\nu(t)$ is bounded. Since $\boldsymbol{Q}$ must be bounded, take $t_0 > 0$ in parts (a) and (c) and $t_0 \ge 0$ in part (b).

Here $\nu(t) = 1/t > 0$ for all $t \ge t_0 > 0$, so $\beta = 1/(\rho t)$ and

$$\int_{t_0}^t \beta(\sigma)\,\mathrm{d}\sigma = \frac{1}{\rho}\ln\left(\frac{t}{t_0}\right) \quad \Rightarrow \quad \|\mathbf{x}(t)\|_2 \le \sqrt{\frac{\rho}{\eta}}\left(\frac{t_0}{t}\right)^{1/(2\rho)}\|\mathbf{x}_0\|_2$$

- *Stable isL:* since $\nu \geq 0$, $\dot{V} \leq 0$ and $\|\mathbf{x}(t)\|_2 \leq \sqrt{\rho/\eta}\|\mathbf{x}_0\|_2$, with $\gamma = \sqrt{\rho/\eta}$ independent of t_0. This is the lecture's positive semi-definite case.
- *Asymptotically stable:* $(t_0/t)^{1/(2\rho)} \to 0$ as $t \to \infty$, so $\|\mathbf{x}(t)\|_2 \to 0$ for every $\mathbf{x}_0$.
- *Uniformity cannot be concluded:* the bound drops below ε only once $t \geq t_0\big(\sqrt{\rho/\eta}/\varepsilon\big)^{2\rho}$, so the $T(\varepsilon, t_0)$ it certifies grows with t_0. Some systems that meet the hypotheses really are not UAS. The test system $\dot{x} = -x/(2t)$ has $\nu = 1/t$ and $x(t) = x_0\sqrt{t_0/t}$, so it needs $T(\varepsilon, t_0) = t_0(\varepsilon^{-2} - 1)$, which grows with t_0, like the lecture example $\boldsymbol{\Phi}(t, t_0) = (1 + t_0)/(1 + t)$.
- *Not exponentially:* $\boldsymbol{Q}(t)$ is positive definite at every instant but not uniformly ($\inf_t \nu(t) = 0$), so the lecture ES test does not apply, and the decay above is only algebraic. The same test system decays like $x_0\sqrt{t_0/t}$, which no envelope $\gamma e^{-\lambda(t-t_0)}$ can bound as $t \to \infty$.

$$\nu(t) = \frac{1}{t} \quad \Rightarrow \quad \text{asymptotically stable (UAS and ES cannot be concluded)} \quad \blacksquare$$

b) Here $\nu(t) = -ce^{-t} < 0$ for all t, so $\boldsymbol{Q}(t)$ is never positive semi-definite and neither lecture test applies. Every instant falls in the $\nu < 0$ case, so $\beta = -(c/\eta)e^{-t}$ and

$$\int_{t_0}^{t} \beta(\sigma)\,\mathrm{d}\sigma = -\frac{c}{\eta}(e^{-t_0} - e^{-t}) \geq -\frac{c}{\eta}e^{-t_0}$$

The integral is bounded below, so by $(*)$, for all $t \geq t_0 \geq 0$,

$$\|\mathbf{x}(t)\|_2 \leq \sqrt{\frac{\rho}{\eta}}\exp\!\left(\frac{c}{2\eta}(e^{-t_0} - e^{-t})\right)\|\mathbf{x}_0\|_2 \leq \sqrt{\frac{\rho}{\eta}}e^{c/(2\eta)}\|\mathbf{x}_0\|_2$$

This is stability isL, with $\gamma = \sqrt{\rho/\eta}e^{c/(2\eta)}$ independent of t_0. V may grow, but the total destabilizing contribution $\int_{t_0}^{\infty}|\nu(\sigma)|\,\mathrm{d}\sigma = ce^{-t_0}$ is finite, so it grows by at most a fixed factor.

Asymptotic stability cannot be concluded. The test system $\dot{x} = (c/2)e^{-t}x$ has $\nu(t) = -ce^{-t}$, yet

$$x(t) = x_0\exp\!\Big(\frac{c}{2}(e^{-t_0} - e^{-t})\Big) \to x_0 e^{ce^{-t_0}/2} \neq 0$$

$$\nu(t) = -ce^{-t} \quad \Rightarrow \quad \text{stable isL (uniformly)} \quad \blacksquare$$

c) Here $\nu(t) = 1/t^2 > 0$, so $\boldsymbol{Q}(t)$ is positive definite at every instant, $\dot{V} \leq 0$, and (as in the lecture theorem) $\|\mathbf{x}(t)\|_2 \leq \sqrt{\rho/\eta}\|\mathbf{x}_0\|_2$: stable isL, with γ independent of t_0. However, ν decays fast enough that its integral converges:

$$\int_{t_0}^{t} \beta(\sigma)\,\mathrm{d}\sigma = \frac{1}{\rho}\left(\frac{1}{t_0} - \frac{1}{t}\right) \to \frac{1}{\rho t_0} < \infty$$

so $(*)$ only gives $\|\mathbf{x}(t)\|_2 \leq \sqrt{\rho/\eta}e^{-(1/t_0 - 1/t)/(2\rho)}\|\mathbf{x}_0\|_2$, which does not tend to zero. This is a real limitation, not a weak bound. The test system $\dot{x} = -x/(2t^2)$ has $\nu = 1/t^2$, and

$$x(t) = x_0\exp\!\left(-\frac{1}{2}\left(\frac{1}{t_0} - \frac{1}{t}\right)\right) \to x_0 e^{-1/(2t_0)} \neq 0$$

So asymptotic stability cannot be concluded. A $\boldsymbol{Q}(t)$ that is positive definite at every instant is not enough when the total dissipation $\int_{t_0}^{\infty}\nu$ is finite.

$$\nu(t) = \frac{1}{t^2} \quad \Rightarrow \quad \text{stable isL (uniformly)} \quad \blacksquare$$

d) Here $\nu(t) = 2 - 3\sin(t)$ ranges over $[-1, 5]$. It is negative when $\sin(t) > 2/3$, i.e. on $(t_1, \pi - t_1) + 2k\pi$ with $t_1 = \arcsin(2/3)$. So $\boldsymbol{Q}(t)$ is not positive semi-definite for part of every period, and neither lecture test applies. Still, ν is positive on average. Over one period, using $\cos(t_1) = \sqrt{5}/3$:

$$I_- = \int_0^{2\pi} \nu_- \, \mathrm{d}t = \int_{t_1}^{\pi - t_1} (3\sin(t) - 2) \, \mathrm{d}t = 2\sqrt{5} - 2(\pi - 2t_1) \approx 1.108$$

$$I_+ = \int_0^{2\pi} \nu_+ \, \mathrm{d}t = \int_0^{2\pi} \nu \, \mathrm{d}t + I_- = 4\pi + I_- \approx 13.674$$

β is 2π-periodic, so its integral over any window of length 2π is

$$\mu = \int_t^{t+2\pi} \beta(\sigma) \, \mathrm{d}\sigma = \frac{I_+}{\rho} - \frac{I_-}{\eta}$$

Write $t - t_0 = 2\pi k + r$ with $0 \le r < 2\pi$. The k full periods contribute $k\mu$, and the leftover piece contributes at least $-I_-/\eta$. So if $\mu > 0$,

$$\int_{t_0}^{t} \beta(\sigma) \, \mathrm{d}\sigma \ge \mu \frac{t - t_0}{2\pi} - \mu - \frac{I_-}{\eta} \quad \Rightarrow \quad \|\mathbf{x}(t)\|_2 \le \sqrt{\frac{\rho}{\eta}} e^{(\mu + I_-/\eta)/2} e^{-\mu(t - t_0)/(4\pi)} \|\mathbf{x}_0\|_2$$

Both constants are independent of t_0, so the system is uniformly exponentially stable with $\lambda = \mu/(4\pi)$ whenever

$$\mu > 0 \quad \Leftrightarrow \quad \frac{\rho}{\eta} < \kappa^* = \frac{I_+}{I_-} = 1 + \frac{4\pi}{2\sqrt{5} - 2\pi + 4\arcsin(2/3)} \approx 12.343$$

In particular this always holds when $\boldsymbol{P}$ is a constant multiple of $\boldsymbol{I}$ ($\rho = \eta$, so $\mu = 4\pi/\rho$). In the boundary case $\rho/\eta = \kappa^*$, $\mu = 0$ and the same window argument gives $\int_{t_0}^{t} \beta \ge -I_-/\eta$. By $(*)$ that is uniform stability isL, with $\gamma = \sqrt{\rho/\eta} e^{I_-/(2\eta)}$. The bound $(*)$ does not decay there, so AS cannot be concluded from this pair.

If $\rho/\eta > \kappa^*$, then $\mu < 0$, the bound grows, and the test is inconclusive. This is not just a loose bound. Take $\boldsymbol{P} = \mathrm{diag}(\rho, \eta)$, $\boldsymbol{Q}(t) = \nu(t)\boldsymbol{I}$, and $\boldsymbol{A}(t) = \boldsymbol{P}^{-1}\left[-\frac{1}{2}\nu(t)\boldsymbol{I} + k(t)\boldsymbol{S}\right]$ with $\boldsymbol{S} = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$. For any bounded scalar $k(t)$, these satisfy the Lyapunov equation with $\min_k \lambda_k(\boldsymbol{Q}(t)) = \nu(t)$. In $\mathbf{y} = \boldsymbol{P}^{1/2}\mathbf{x}$ the dynamics are $\dot{\mathbf{y}} = \left[-\frac{1}{2}\nu\boldsymbol{P}^{-1} + k/\sqrt{\rho\eta}\, \boldsymbol{S}\right]\mathbf{y}$, so the $k(t)$ term only rotates $\mathbf{y}$ and leaves $V = \|\mathbf{y}\|_2^2$ unchanged. Start at $t_0 = 0$ with $\mathbf{x}_0$ on the ρ-axis ($\nu(0) = 2 > 0$). Let $k(t) = \pm(\pi/2)\sqrt{\rho\eta}/\delta$ on windows of length δ at each zero crossing of ν, and 0 elsewhere, which makes a quarter turn at each crossing. Then the state lies along the η-axis while $\nu < 0$, where V grows at rate $|\nu|/\eta$. It lies along the ρ-axis while $\nu > 0$, where V decays only at rate ν/ρ. Each period multiplies V by $e^{-\mu}$, up to an error that vanishes as $\delta \to 0$. For any $\rho/\eta > \kappa^*$ a small enough δ makes that factor exceed 1, so this system is unstable.

$$\nu(t) = 2 - 3\sin(t) \quad \Rightarrow \quad \begin{cases} \text{uniformly exponentially stable} & \text{if } \rho/\eta < \kappa^* \approx 12.343 \\ \text{stable isL (uniformly)} & \text{if } \rho/\eta = \kappa^* \\ \text{inconclusive} & \text{if } \rho/\eta > \kappa^* \end{cases} \qquad \blacksquare$$

Problem 7

Generalize your observations from Problem 6. What general conditions on $\nu(t)$ would allow you to conclude isL stability? Asymptotic stability? Exponential stability?

NOTE: This question is asking you to go beyond the comparatively straightforward results discussed in lecture; for example to demonstrate formally that stability can sometimes be concluded if $\nu(t) < 0$ for some or even all $t \geq t_0$, etc.

All four cases of Problem 6 follow from one bound. Assume $\boldsymbol{P}(t)$ is bounded and uniformly positive definite ($\eta \boldsymbol{I} \leq \boldsymbol{P}(t) \leq \rho \boldsymbol{I}$) and $\boldsymbol{Q}(t)$ is bounded ($\|\boldsymbol{Q}(t)\|_i \leq q$, so $\nu(t) \geq -q$). With $V = \mathbf{x}^\mathsf{T}\boldsymbol{P}\mathbf{x}$, the argument at the start of Problem 6a gives

$$\dot{V} \leq -\beta(t)V, \quad \beta(t) = \frac{\nu_+(t)}{\rho} - \frac{\nu_-(t)}{\eta} \quad \Rightarrow \quad \|\mathbf{x}(t)\|_2 \leq \sqrt{\frac{\rho}{\eta}} \exp\Big(-\frac{1}{2}B(t,t_0)\Big)\|\mathbf{x}_0\|_2, \quad B(t,t_0) = \int_{t_0}^{t} \beta(\sigma)\,\mathrm{d}\sigma$$

where $\nu_\pm = \max\{\pm\nu, 0\}$. The positive part of ν is weighted by $1/\rho$ and the negative part by $1/\eta$. Where $\nu < 0$, the state can sit in the direction where $\boldsymbol{P}$ is smallest, making $\|\mathbf{x}\|_2^2$ as large as V/η. So what this $(\boldsymbol{P}, \boldsymbol{Q})$ pair can certify is decided by the running integral $B(t, t_0)$, not by the sign of ν at each instant:

- *Stable isL* if B is bounded below: $B(t,t_0) \geq -M(t_0) > -\infty$ for all $t \geq t_0$. Then $\gamma(t_0) = \sqrt{\rho/\eta}e^{M(t_0)/2}$, and the stability is uniform if M does not depend on t_0. A sufficient condition on ν alone is integrability of its negative part:

  $$\int_{t_0}^{\infty} \nu_-(\sigma)\,\mathrm{d}\sigma < \infty \quad \Rightarrow \quad \|\mathbf{x}(t)\|_2 \leq \sqrt{\frac{\rho}{\eta}} \exp\left(\frac{1}{2\eta}\int_{t_0}^{\infty} \nu_-(\sigma)\,\mathrm{d}\sigma\right)\|\mathbf{x}_0\|_2$$

  since $B(t,t_0) \geq -\frac{1}{\eta}\int_{t_0}^{t} \nu_-$. This stability is automatically uniform over all $t_0 \geq t_s$ (for any fixed earliest start time t_s), because $\int_{t_0}^{\infty} \nu_- \leq \int_{t_s}^{\infty} \nu_-$. So stability can be concluded even if $\nu(t) < 0$ for some or all $t \geq t_0$, as long as the total negative contribution is finite (6b). The lecture condition, $\boldsymbol{Q}$ positive semi-definite ($\nu \geq 0$), is the special case $\nu_- \equiv 0$, $M = 0$. If $\nu(t) < 0$ for all t, then $B \leq 0$, so stability isL is the most this pair can certify; AS and ES cannot be concluded.

- *Asymptotically stable* if $B(t,t_0) \to +\infty$ as $t \to \infty$, since then $\|\mathbf{x}(t)\|_2 \to 0$ for every $\mathbf{x}_0$. This already includes stability isL: B is continuous with $B(t_0,t_0) = 0$, so a B that tends to $+\infty$ is bounded below. A sufficient condition on ν alone is

  $$\int_{t_0}^{\infty} \nu_-(\sigma)\,\mathrm{d}\sigma < \infty \quad \text{and} \quad \int_{t_0}^{\infty} \nu_+(\sigma)\,\mathrm{d}\sigma = \infty$$

  The divergence matters. $\nu = 1/t$ (6a) and $\nu = 1/t^2$ (6c) are both positive, but only the first has a divergent integral, and only the first gives AS. If B diverges more slowly for later t_0 (as $\ln(t/t_0)$ does), the time the bound needs to reach a given ε grows with t_0, so uniformity cannot be concluded from this bound.

- *Exponentially stable* if B grows at least linearly,

  $$B(t,t_0) \geq 2\lambda(t - t_0) - M \quad \forall t \geq t_0, \quad \text{for some } \lambda > 0, M < \infty$$

  which gives $\|\mathbf{x}(t)\|_2 \leq \sqrt{\rho/\eta}e^{M/2}e^{-\lambda(t-t_0)}\|\mathbf{x}_0\|_2$. If λ and M do not depend on t_0, it is UES. A convenient sufficient condition is a uniformly positive windowed integral: for some $T > 0$ and $\mu > 0$,

  $$\int_{t}^{t+T} \beta(\sigma)\,\mathrm{d}\sigma = \int_{t}^{t+T} \left[\frac{\nu_+(\sigma)}{\rho} - \frac{\nu_-(\sigma)}{\eta}\right]\mathrm{d}\sigma \geq \mu \quad \forall t \geq t_0$$

Chop $[t_0, t]$ into windows of length T, as in the lecture proof that UAS implies UES. Each full window contributes at least μ, and the leftover piece at least $-qT/\eta$, so $\lambda = \mu/(2T)$ and $M = \mu + qT/\eta$, independent of t_0: UES. The lecture case ($\boldsymbol{Q}$ uniformly positive definite, $\nu(t) \geq \nu_{\min} > 0$) is the special case $\mu = \nu_{\min} T/\rho$, which recovers $\lambda = \nu_{\min}/(2\rho)$. 6d satisfies the condition with $\nu < 0$ on part of every period, provided $\rho/\eta < \kappa^* \approx 12.343$. A version that uses ν alone and works for every ρ/η: if $\int_{t_0}^{\infty} \nu_- = N < \infty$ and $\int_t^{t+T} \nu_+ \geq \mu' > 0$ for all t, then $B(t, t_0) \geq (\mu'/\rho)((t - t_0)/T - 1) - N/\eta$. Since $N \leq \int_{t_s}^{\infty} \nu_-$ for every $t_0 \geq t_s$, the constant $M = \mu'/\rho + N/\eta$ is uniform, and this is again UES.

These conditions are sufficient, not necessary. When they fail, the result is inconclusive. The bound comes from one particular $(\boldsymbol{P}, \boldsymbol{Q})$ pair, and by the converse theorem a UES system has infinitely many other pairs, some of which may pass. The conditions also cannot be weakened using only $\nu(t)$. For the scalar systems $\dot{x} = a(t)x$ with constant $P = p$ (so $\rho = \eta = p$), $\dot{V} = -\nu x^2 = -(\nu/p)V$ holds with equality. The bound is then exact, and each condition on B is also necessary for that family, as the test systems in Problem 6 showed. The different weights $1/\rho$ and $1/\eta$ matter only when ν changes sign, which is why the conditioning ρ/η of $\boldsymbol{P}$ entered 6d. The rotating construction there shows that the $1/\eta$ weight on ν_- cannot be improved when $\rho \neq \eta$. In summary, with $B(t, t_0) = \int_{t_0}^{t} [\nu_+/\rho - \nu_-/\eta] \, \mathrm{d}\sigma$:

$$\begin{aligned}
\text{stable isL} \Longleftarrow \quad & \inf_{t \geq t_0} B(t, t_0) > -\infty \quad \left(\Longleftarrow \int_{t_0}^{\infty} \nu_- < \infty \right) \\
\text{asymptotically stable} \Longleftarrow \quad & B(t, t_0) \to \infty \\
\text{exponentially stable} \Longleftarrow \quad & B(t, t_0) \geq 2\lambda(t - t_0) - M, \quad \text{uniformly (UES) if} \int_t^{t+T} \beta \geq \mu > 0 \ \forall t
\end{aligned} \qquad \blacksquare$$

Code

./src/index.m Matlab

```
%% ENAE 641 - Homework 3
% Vai Srivastava

% clear history
clear;
clc;
close all;

% resolve paths from this file, so the script runs from any working folder
root = fileparts(fileparts(mfilename("fullpath")));
textDir = fullfile(root, "outputs", "text");
figDir = fullfile(root, "outputs", "figures");
if ~isfolder(textDir), mkdir(textDir); end
if ~isfolder(figDir), mkdir(figDir); end

% load provided data
load(fullfile(root, "data", "F16.mat"), "A", "B", "C");
n = size(A, 1);
stateNames = ["speed"; "angle of attack"; "pitch angle"; "pitch rate"];
sys = ss(A, B, C, 0);

% categorical series colours (fixed order) and chart ink
colors = ["#2a78d6", "#eb6834", "#1baf7a", "#eda100"];
muted = "#52514e";

%% Problem 1a: real block modal form
[V, D] = eig(A);
lambda = diag(D);

% keep the +j*omega member of each conjugate pair, ordered slow -> fast
% (phugoid, then short period)
keep = imag(lambda) > 0;
lambdaModes = lambda(keep);
vModes = V(:, keep);
[~, order] = sort(abs(lambdaModes));
lambdaModes = lambdaModes(order);
vModes = vModes(:, order);

% scale each eigenvector so that its largest entry is exactly 1
for k = 1:2
    [~, iMax] = max(abs(vModes(:, k)));
    vModes(:, k) = vModes(:, k) / vModes(iMax, k);
end
vPh = vModes(:, 1);
```

```
vSp = vModes(:, 2);

% x = Pm z, with Pm = [Re(v_ph), Im(v_ph), Re(v_sp), Im(v_sp)]
Pm = [real(vPh), imag(vPh), real(vSp), imag(vSp)];
Ap = Pm \ A * Pm;
Ap(abs(Ap) < 1e-12) = 0;
Bp = Pm \ B;

sigma = real(lambdaModes);
omega = imag(lambdaModes);
wn = abs(lambdaModes);
zeta = -sigma ./ wn;
period = 2 * pi ./ omega;
modes = table(["phugoid"; "short period"], sigma, omega, wn, zeta, period, ...
    VariableNames=["mode", "sigma", "omega", "omega_n", "zeta", "period_s"]);

report(textDir, "s01a", join([
    "eigenvalues of A:"
    show(lambda)
    ""
    "modes (lambda = sigma +/- j omega):"
    show(modes)
    ""
    "max Re(lambda) = " + num2str(max(real(lambda)), 6) + " < 0  ->  asymptotically
    stable"
    ""
    "Pm = [Re(v_ph), Im(v_ph), Re(v_sp), Im(v_sp)]:"
    show(Pm)
    ""
    "A_p = Pm^-1 A Pm:"
    show(Ap)
], newline));

%% Problem 1b: single-mode initial conditions
polarPh = table(stateNames, abs(vPh), rad2deg(angle(vPh)), VariableNames=["state", "r",
"phi_deg"]);
polarSp = table(stateNames, abs(vSp), rad2deg(angle(vSp)), VariableNames=["state", "r",
"phi_deg"]);

% x0 in span{Re(v), Im(v)} excites only that mode; take x0 = Re(v)
x0Ph = real(vPh);
x0Sp = real(vSp);

tPh = linspace(0, 400, 2001)';
tSp = linspace(0, 6, 1201)';
[~, tPh, xPh] = initial(sys, x0Ph, tPh);
[~, tSp, xSp] = initial(sys, x0Sp, tSp);
```

```
zPh = (Pm \ xPh')';
zSp = (Pm \ xSp')';

report(textDir, "s01b", join([
    "phugoid eigenvector v_ph (polar form, relative to speed):"
    show(polarPh)
    ""
    "short-period eigenvector v_sp (polar form, relative to pitch rate):"
    show(polarSp)
    ""
    "phugoid x0 = Re(v_ph):"
    show(x0Ph')
    "z0 = Pm^-1 x0:"
    show((Pm \ x0Ph)')
    "max |z3|, |z4| over the phugoid response:  " + num2str(max(abs(zPh(:, 3:4)), [], "all"), 3)
    ""
    "short-period x0 = Re(v_sp):"
    show(x0Sp')
    "z0 = Pm^-1 x0:"
    show((Pm \ x0Sp)')
    "max |z1|, |z2| over the short-period response:  " + num2str(max(abs(zSp(:, 1:2)), [], "all"), 3)
], newline));

fig = figure(Visible="off", Units="centimeters", Position=[0, 0, 18, 12]);
layout = tiledlayout(fig, 2, 2, TileSpacing="compact", Padding="compact");
cases = {tPh, xPh, zPh, "phugoid: x_0 = Re\{v_{ph}\}"; tSp, xSp, zSp, "short period: x_0 = Re\{v_{sp}\}"};
for c = 1:2
    [t, x, z, label] = cases{c, :};

    ax = nexttile(layout, c);
    hold(ax, "on");
    for i = 1:n
        plot(ax, t, x(:, i), Color=colors(i), LineWidth=1.5);
    end
    styleAxes(ax, muted);
    title(ax, label);
    ylabel(ax, "state x_i(t)");
    if c == 1
        legend(ax, stateNames, Location="northeast", Box="off");
    end

    ax = nexttile(layout, c + 2);
    hold(ax, "on");
    for i = 1:n
```

```
133         plot(ax, t, z(:, i), Color=colors(i), LineWidth=1.5);
134     end
135     styleAxes(ax, muted);
136     xlabel(ax, "time t (s)");
137     ylabel(ax, "modal coordinate z_i(t)");
138     if c == 1
139         legend(ax, "z_" + (1:n), Location="northeast", Box="off");
140     end
141 end
142 saveFigure(fig, figDir, "s01b");
143
144 %% Problem 1c: modal B matrix
145 % rows: phugoid block (z1, z2), short-period block (z3, z4); columns: throttle, elevator
146 blocks = {1:2, 3:4};
147 blockNorm = zeros(2, 2);
148 projNorm = zeros(2, 2);
149 for k = 1:2
150     for j = 1:2
151         blockNorm(k, j) = norm(Bp(blocks{k}, j));
152         % part of b_j lying in mode k's plane (independent of eigenvector scaling)
153         projNorm(k, j) = norm(Pm(:, blocks{k}) * Bp(blocks{k}, j));
154     end
155 end
156 inputs = ["throttle", "elevator"];
157 blockTable = array2table(blockNorm, VariableNames=inputs, RowNames=["phugoid", "short
    period"]);
158 projTable = array2table(projNorm, VariableNames=inputs, RowNames=["phugoid", "short
    period"]);
159
160 report(textDir, "s01c", join([
161     "B_p = Pm^-1 B (columns: throttle, elevator):"
162     show(Bp)
163     ""
164     "modal-block norms ||B_p(block, j)||:"
165     show(blockTable)
166     "ratios (larger / smaller): throttle " + num2str(blockNorm(1, 1) / blockNorm(2, 1),
    3) ...
167         + ", elevator " + num2str(blockNorm(2, 2) / blockNorm(1, 2), 3)
168     ""
169     "projection of each input direction onto each mode's plane, ||Pm_k B_p,k(:, j)||:"
170     show(projTable)
171     "ratios (larger / smaller): throttle " + num2str(projNorm(1, 1) / projNorm(2, 1),
    3) ...
172         + ", elevator " + num2str(projNorm(2, 2) / projNorm(1, 2), 3)
173 ], newline));
174
175 %% Problem 4a: Lyapunov matrix
```

```
% lyap(A, Q) solves A*X + X*A' + Q = 0, so pass A' to solve A'*P + P*A + Q = 0
Q = eye(n);
P = lyap(A', Q);
eigP = eig(P);
[~, cholFlag] = chol(P);

report(textDir, "s04a", join([
    "P = lyap(A', I):"
    show(P)
    ""
    "eigenvalues of P:"
    show(eigP')
    "min eigenvalue = " + num2str(min(eigP), 6) + " > 0,  chol flag = " + cholFlag ...
        + "  ->  P is positive definite"
], newline));

%% Problem 4b: exponential convergence bound
rho = max(eigP);
eta = min(eigP);
nu = min(eig(Q));
gamma = sqrt(rho / eta);
lambdaBound = nu / (2 * rho);
slowestRate = min(-real(eig(A)));

tB = (0:0.01:600)';
gainA = arrayfun(@(t) norm(expm(A * t)), tB);
bound = gamma * exp(-lambdaBound * tB);
[peakGain, iPeak] = max(gainA);

% where the conservatism comes from
phPlane = orth(Pm(:, 1:2));
rhoPh = max(eig(phPlane' * P * phPlane));
reV = Pm(:, 1);
imV = Pm(:, 2);
phAngle = acosd(reV' * imV / (norm(reV) * norm(imV)));
minM = max(gainA .* exp(lambdaBound * tB));

report(textDir, "s04b", join([
    "rho = lambda_max(P) = " + num2str(rho, 6)
    "eta = lambda_min(P) = " + num2str(eta, 6)
    "nu  = lambda_min(Q) = " + num2str(nu, 6)
    "gamma  = sqrt(rho / eta) = " + num2str(gamma, 6)
    "lambda = nu / (2 rho)   = " + num2str(lambdaBound, 6) + " 1/s"
    ""
    "slowest decay rate of A, min |Re(lambda)| = " + num2str(slowestRate, 6) + " 1/s"
    "max_t ||e^(At)||_2 = " + num2str(peakGain, 6) + " at t = " + num2str(tB(iPeak), 4) +
    " s"
```

```
    "bound >= ||e^(At)||_2 on the whole grid: " + string(all(bound >= gainA))
    ""
    "lambda_max of P restricted to the phugoid plane = " + num2str(rhoPh, 4) ...
        + "  (1 / (2 |sigma_1|) = " + num2str(1 / (2 * abs(sigma(1))), 4) + ")"
    "|Re(v_ph)| = " + num2str(norm(reV), 3) + ", |Im(v_ph)| = " + num2str(norm(imV), 3) ...
        + ", angle between them = " + num2str(phAngle, 3) + " deg"
    "smallest M with ||e^(At)||_2 <= M e^(-lambda t): " + num2str(minM, 4)
    "|sigma_3| / |sigma_1| = " + num2str(sigma(2) / sigma(1), 4)
    "modal bound: ||x(t)|| <= cond(Pm) e^(sigma_1 t) ||x0||, cond(Pm) = " + num2str(cond(Pm), 4)
], newline));

fig = figure(Visible="off", Units="centimeters", Position=[0, 0, 16, 8]);
ax = axes(fig);
hold(ax, "on");
plot(ax, tB, gainA, Color=colors(1), LineWidth=1.5);
plot(ax, tB, bound, Color=colors(2), LineWidth=1.5);
ax.YScale = "log";
styleAxes(ax, muted);
xlabel(ax, "time t (s)");
ylabel(ax, "||x(t)||_2 / ||x_0||_2");
legend(ax, ["worst case ||e^{At}||_2", "Lyapunov bound \gamma e^{-\lambda t}"], Location="northeast", Box="off");
saveFigure(fig, figDir, "s04b");

%% Problem 5a: eigenvalues of A + A'
M = A + A';
eigM = eig(M);

report(textDir, "s05a", join([
    "eigenvalues of A + A':"
    show(eigM')
    "number of positive eigenvalues: " + sum(eigM > 0)
], newline));

%% Problem 5b: ||x||^2 vs x'Px along solutions
% x0 = w: ||x||^2 grows at t = 0 (largest eigenvalue of A + A')
[W, E] = eig(M);
[~, iTop] = max(diag(E));
w = W(:, iTop);
growth0 = w' * M * w;

% x0 = xw: worst case at the peak of ||e^(At)||_2 (top right singular vector)
[~, ~, Vsvd] = svd(expm(A * tB(iPeak)));
xw = Vsvd(:, 1);

```

```
t5 = (0:0.005:60)';
ics = [w, xw];
normRatio = zeros(numel(t5), 2);
vRatio = zeros(numel(t5), 2);
for c = 1:2
    [~, ~, x5] = initial(sys, ics(:, c), t5);
    normRatio(:, c) = vecnorm(x5, 2, 2) / norm(ics(:, c));
    vRatio(:, c) = sum((x5 * P) .* x5, 2) / (ics(:, c)' * P * ics(:, c));
end
[peakNorm, iPk] = max(normRatio);
iBack = find(normRatio(:, 1) < 1 & t5 > t5(iPk(1)), 1);

report(textDir, "s05b", join([
    "x0 = w, the unit eigenvector of A + A' for its largest eigenvalue:"
    show(w')
    "d/dt ||x||^2 at t = 0: w'(A + A')w = " + num2str(growth0, 6) + " > 0"
    "max ||x(t)|| / ||x0|| = " + num2str(peakNorm(1), 6) + " at t = " +
    num2str(t5(iPk(1)), 4) ...
        + " s (back below 1 at t = " + num2str(t5(iBack), 4) + " s)"
    ""
    "x0 = x_w, the unit vector maximizing ||e^(At)x0|| at t = " + num2str(tB(iPeak), 4) +
    " s:"
    show(xw')
    "max ||x(t)|| / ||x0|| = " + num2str(peakNorm(2), 6) + " at t = " +
    num2str(t5(iPk(2)), 4) + " s"
    ""
    "V = x'Px nonincreasing along both solutions: " + string(all(diff(vRatio) <= 0,
    "all"))
], newline));

fig = figure(Visible="off", Units="centimeters", Position=[0, 0, 18, 7.5]);
layout = tiledlayout(fig, 1, 3, TileSpacing="compact", Padding="compact");
% norm over 60 s, the same norm over the first second, and V over 60 s
panels = {normRatio, 60, "||x(t)||_2 / ||x_0||_2"; normRatio, 1, "||x(t)||_2 / ||x_0||_2,
first 1 s"; ...
    vRatio, 60, "V(t) / V(0),  V = x^T P x"};
for c = 1:3
    ax = nexttile(layout);
    hold(ax, "on");
    yline(ax, 1, Color="#c3c2b7", LineWidth=0.8, HandleVisibility="off");
    inWindow = t5 <= panels{c, 2};
    for i = 1:2
        plot(ax, t5(inWindow), panels{c, 1}(inWindow, i), Color=colors(i), LineWidth=1.5);
    end
    styleAxes(ax, muted);
    xlabel(ax, "time t (s)");
    title(ax, panels{c, 3});
```

```
    if c == 1
        legend(ax, ["x_0 = w", "x_0 = x_w"], Location="southwest", Box="off");
    end
end
saveFigure(fig, figDir, "s05b");

%% Helper functions
function out = show(value)
    % display text for a value, without trailing blank lines
    out = strip(formattedDisplayText(value, NumericFormat="shortg", ...
        LineSpacing="compact", SuppressMarkup=true), "right");
end

function report(dir, name, txt)
    % print a result and save it for the report
    fprintf("\n------\n %s\n------\n%s\n", name, txt);
    writelines(txt, fullfile(dir, name + ".txt"));
end

function styleAxes(ax, ink)
    % recessive axes and grid
    ax.Box = "off";
    ax.FontSize = 9;
    ax.TickDir = "out";
    ax.LineWidth = 0.6;
    ax.XColor = ink;
    ax.YColor = ink;
    ax.XGrid = "on";
    ax.YGrid = "on";
    ax.GridColor = "#c3c2b7";
    ax.GridAlpha = 0.5;
end

function saveFigure(fig, dir, name)
    % export a figure for the report
    exportgraphics(fig, fullfile(dir, name + ".png"), Resolution=200);
    close(fig);
end
```